Adyasha Mohanty

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Summary

Electronics & Computer Science undergraduate focused on AI/ML, deep learning, generative and agentic AI, and MLOps. Built end-to-end ML systems and interested in data analytics.

Education

- **KIIT University** — Bhubaneswar, India
  B.Tech – Electronics and Computer Science, CGPA: 7.00/10.00 — Aug 2023 – Present
  - Coursework: DSA, Operating Systems, OOP, Databases, Computer Networks, Cloud Computing, Cryptography, Network Security, Computer Vision

Technical Skills

- **Languages & CS Fundamentals:** Python, Java, SQL, C++, Bash, DSA, OOP
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, CNNs, Transformers, Computer Vision, NLP, Time-Series, Explainable AI (SHAP, Grad-CAM)
- **Generative & Agentic AI:** LLMs, Prompt Engineering, RAG, LangChain, LlamaIndex, LangGraph, AI Agents & Tool Calling, Embeddings, Vector Databases (FAISS, ChromaDB), Hugging Face, Fine-Tuning (LoRA/PEFT)
- **MLOps & Cloud:** MLflow, DVC, Docker, Kubernetes, CI/CD (GitHub Actions), Airflow, FastAPI, AWS, GCP, Git
- **Data & Analytics:** Pandas, NumPy, Power BI, Tableau, A/B Testing, Statistics, EDA
- **Software & Data Engineering:** REST APIs, Microservices, Kafka, PostgreSQL, MongoDB, Redis, React

Projects

- **GastricSentinel: AI-Powered Gastric Cancer Detection System** — Dec 2025 – Mar 2026
  Python, PyTorch, Scikit-learn, Flask, JavaScript | github.com/adyashaexe/GastricSentinel
  - Trained a multimodal deep learning pipeline on 245K+ histopathology images and ∼450 TCGA-STAD genomic cases, achieving 64% binary classification accuracy on noisy and imbalanced medical data.
  - Performed preprocessing, feature extraction, and evaluation, with Grad-CAM and SHAP for interpretability.
  - Deployed the model via a Flask REST API and JavaScript dashboard for real-time inference.
- **AegisNet: AI-Based Network Intrusion Detection System** — Feb 2026 – Mar 2026
  Python, Scikit-learn, Scapy, Flask, JavaScript | github.com/adyashaexe/AegisNet
  - Built a real-time IDS on 2.8M+ CIC-IDS-2017 flows (13 attack classes) using Scapy and Scikit-learn.
  - Engineered a 78-feature pipeline with SMOTE oversampling to improve recall on rare attack classes.
  - Designed a 4-stage pipeline (capture, processing, classification, alerting) with a Flask backend and dashboard.
- **HardwareTrojanML: Hardware Trojan Detection in FPGA AES-128** — Jul 2026 – Present
  Python, PyTorch, Scikit-learn, Verilog, Vivado, Arduino | github.com/adyashaexe/HardwareTrojanML
  - Built a power-trace dataset (clean AES-128 and 3 Trojan variants) on a Basys3 FPGA with an INA219 sensor.
  - Windowed and z-score normalized the traces and trained Random Forest, SVM, and PyTorch 1D-CNN models; the 1D-CNN performed best at 77% four-class accuracy (Random Forest: 68%).
  - Evaluated with per-class recall and clean-vs-Trojan FNR/FPR, and documented limitations and next steps.

Certifications

- **Generative AI with LangChain and Hugging Face** – Udemy, 70 hours (Certificate) — Oct 2026
- **Data Science, ML, DL and NLP Bootcamp** – Udemy, 101.5 hours (Certificate) — Oct 2026
- **SAP BTP Developer** – ExcelR & KIIT University, 45 hours (Certificate) — Mar 2026
- **Web Development** – Udemy, 97 hours (Certificate) — Dec 2025

Activities

- **Open Source Contributor**, GirlScript Summer of Code (GSSoC) 2026 — Apr 2026 – Jul 2026
- **Participant**, McKinsey Forward Program (10-week program) — Oct 2026 – Present
- **Graphic Designer and Content Writer**, NSS Lakshya, School of Electronics — Aug 2025 – May 2026